

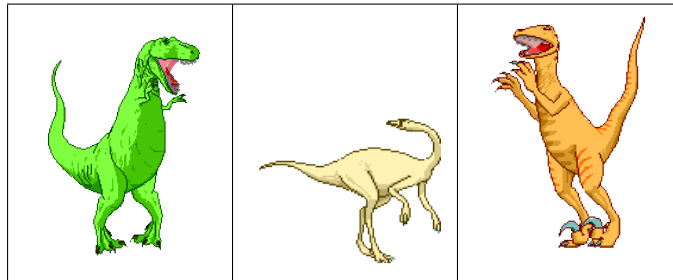
Learning target AD8, version 1

Model and analyze scenarios using applied optimization.

Hints: Draw three pictures; label variables and constants; write a constraint equation; write an objective function that just has one variable; find any endpoints; do the finding extrema recipe.

You are a dinosaur rancher, and you need to build pens for your three dinosaurs. You decide to build an electric fence around a rectangular region with total area of 25 square kilometers, and then separate this region into three equal-size rectangular pens with two more strips of electric fence. What's the minimum length of electric fencing you need to do this?

Here is one of your three pictures.



The minimum length of fence is $\underbrace{\hspace{1cm}}_{\text{number}} \underbrace{\hspace{1cm}}_{\text{units}}$,
 and the dimensions of the dinosaur pen that make this work are $\underbrace{\hspace{1cm}}_{\text{number}} \underbrace{\hspace{1cm}}_{\text{units}} \times \underbrace{\hspace{1cm}}_{\text{number}} \underbrace{\hspace{1cm}}_{\text{units}}$.

Learning target AD8, version 2

Model and analyze scenarios using applied optimization.

If you take a regular 8.5" x 11" sheet of paper and cut squares out of the corners, you can fold up the flaps on all four sides to make a box (without a lid). How large should you cut the squares so that the resulting box has maximum volume?

Hints: Draw three pictures; label variables and constants; write a constraint equation; write an objective function that just has one variable; find any endpoints; do the finding extrema recipe.

The maximum volume is $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{1.5cm}}_{\text{units}},$
 and the dimensions of the box that make this work are $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{1.5cm}}_{\text{units}} \times \underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{1.5cm}}_{\text{units}} \times \underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{1.5cm}}_{\text{units}}.$